



Seeing Through Deepfakes: A Detection System Using Transparency Signals in Generative AI

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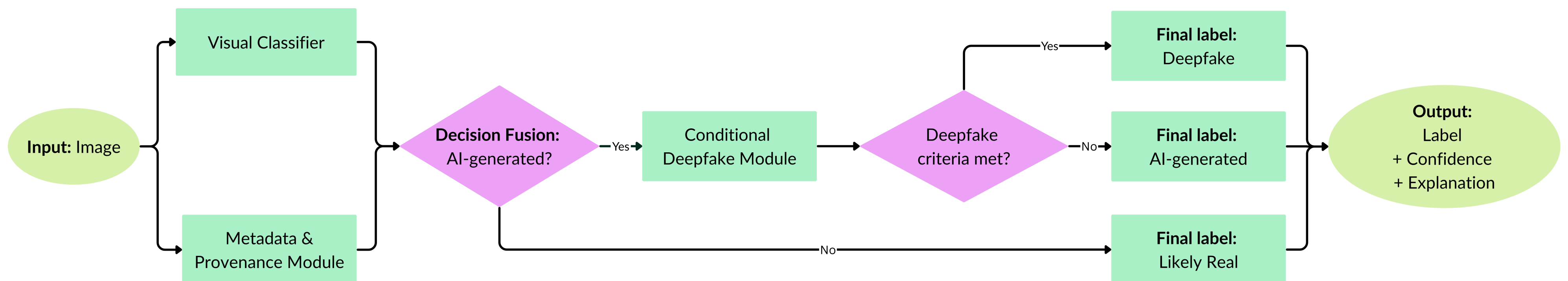
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INTRODUCTION

Generative AI has made it increasingly easy to create realistic synthetic images and deepfakes, prompting apprehensions around misinformation, impersonation and the **manipulation of digital media**. With such content increasingly difficult to distinguish from genuine imagery, regulators and organisations require **effective tools** to underpin transparency, trust and evidence-based assessment. This placement with the **Malta Digital Innovation Authority** developed a **proof-of-concept transparency support tool for AI-generated and manipulated images**. The project was motivated by the EU AI Act's broad definition of deepfakes as AI-generated or manipulated content that may resemble persons, objects, places, entities or events and falsely appear authentic [1]. The project explored a **multi-signal methodology**, rather than relying on a single detector, integrating metadata, provenance indicators and visual classification into an **interpretable decision-support pipeline**.

METHODOLOGY



RESULTS AND EVALUATION

The prototype was tested as a **modular decision-support pipeline** instead of a standalone detector. The metadata module performed best when explicit provenance, software or C2PA-style traces were present, and **missing metadata was treated as uncertainty**, rather than as evidence of authenticity. The visual classifier also improved after fine-tuning, moving from a base model that strongly favoured the real class to a **more balanced real/AI-generated decision boundary** on the controlled test data.

The fused system correctly classified all six real images as real and both non-deepfake AI-generated images as AI-generated, showing that **transparency-based signals, when combined, can reduce false positives and support more interpretable detection decisions**. However, the main limitation appeared in deepfake-style images, where all four were misclassified as real, a **known weak point for visual classifiers rather than a failure of the fusion approach itself**. This was likely caused by the visual classifier assigning very low AI probabilities, even when metadata indicated a possible AI origin. Overall, the evaluation **shows the value of combining transparency signals** but also demonstrates that fusion must be **carefully calibrated** when one component is confidently wrong. The results should therefore be treated as **indicative, not conclusive**.

CONCLUSIONS AND FUTURE WORK

Future work should focus on **strengthening the visual classifier** through broader datasets, newer generators, post-processing tests and better calibration across difficult deepfake-style images. The framework could also be **extended with watermark detection, perceptual hashing and stronger provenance verification**, provided these signals can be evaluated with publicly accessible tools. Further development should **expand semantic analysis beyond faces and places**, reflecting the EU AI Act's wider reference to persons, objects, places, entities and events.

Ultimately, this work highlights the importance of combining multiple analytical perspectives when evaluating synthetic media. The project shows that **trustworthy synthetic-media assessment cannot depend on a single detector**, the **real challenge is learning how imperfect signals** can still reveal a convincing illusion.

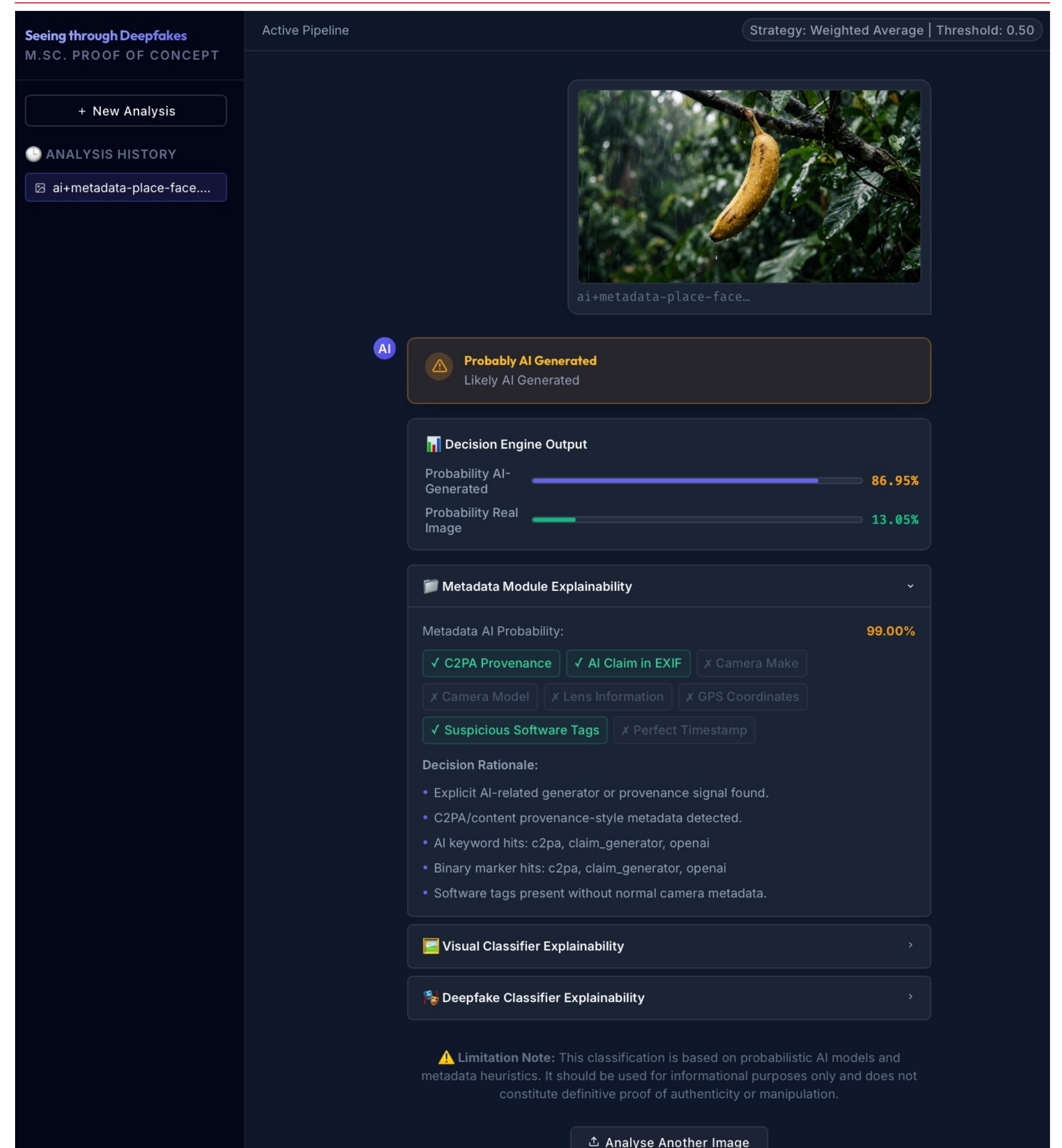
AIM

This work placement focused on the design, implementation and evaluation of a proof-of-concept system to **detect AI-generated or manipulated images using a multi-signal approach**. The prototype combined metadata, provenance information and visual classification into a modular pipeline, providing outputs that were **designed to be interpretable** and useful to expert and non-expert users.

To do this, the placement worked through **three connected objectives**:

1. Analyse relevant **regulatory and policy developments** concerning **transparency for AI-generated and manipulated content** and assess how far **existing frameworks** address the risks involved.
2. Develop a **modular prototype** integrating metadata and provenance analysis, model-based visual classification and decision-level fusion.
3. Evaluate the prototype outputs with respect to detection **performance, interpretability and practical limitations** in a regulatory decision-support context.

PROTOTYPE OUTPUT EXAMPLE



REFERENCES

- [1] European Union, 'Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act)', Jul. 2024, [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2024/1689/>